

● MEDIUM

● ALGEBRA

Systems of two linear equations in two variables

02

10 questions · ~15 min

Q1

ALGEBRA

$$\begin{aligned}y &= 2x - 3 \\ 3y &= 5x\end{aligned}$$

In the solution to the system of equations above, what is the value of y ?

- A. -15
- B. -9
- C. 9
- D. 15

Q2

ALGEBRA

$$y = 6x + 16$$

$$-7x - y = 36$$

What is the solution x, y to the given system of equations?

- A. -4, -8
- B. $-\frac{20}{13}, -\frac{80}{13}$
- C. 4, 40
- D. 20, 136

Q3

ALGEBRA

A proposal for a new library was included on an election ballot. A radio show stated that 3 times as many people voted in favor of the proposal as people who voted against it. A social media post reported that 15,000 more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal?

- A. 7,500
- B. 15,000
- C. 22,500
- D. 45,000

$$y = 2x + 10$$

$$y = 2x - 1$$

At how many points do the graphs of the given equations intersect in the xy -plane?

- A. Zero
- B. Exactly one
- C. Exactly two
- D. Infinitely many

Q5

GRID-IN ALGEBRA

$$2a + 8b = 198$$

$$2a + 4b = 98$$

The solution to the given system of equations is a, b . What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The sum of a number x and 7 is twice as large as a number y . The number y is 3 less than the number x . Which system of equations describes this situation?

A. $x + 7 = 2y$
 $y = x - 3$

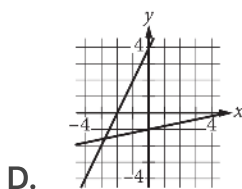
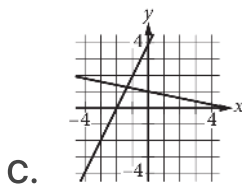
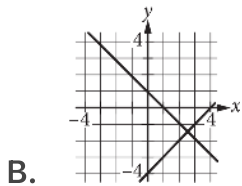
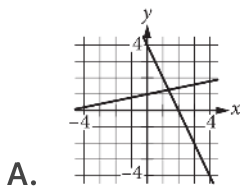
B. $x + 7 = 2y$
 $y = 3 - x$

C. $2x + 7 = y$
 $y = x - 3$

D. $2x + 7 = y$
 $y = 3 - x$

$$\begin{aligned}x + 5y &= 5 \\ 2x - y &= -4\end{aligned}$$

Which of the following graphs in the xy -plane could be used to solve the system of equations above?



A group of 202 people went on an overnight camping trip, taking 60 tents with them. Some of the tents held 2 people each, and the rest held 4 people each. Assuming all the tents were filled to capacity and every person got to sleep in a tent, exactly how many of the tents were 2-person tents?

- A. 30
- B. 20
- C. 19
- D. 18

A bus traveled on the highway and on local roads to complete a trip of 160 miles. The trip took 4 hours. The bus traveled at an average speed of 55 miles per hour mph on the highway and an average speed of 25 mph on local roads. If x is the time, in hours, the bus traveled on the highway and y is the time, in hours, it traveled on local roads, which system of equations represents this situation?

- A. $55x + 25y = 4$
 $x + y = 160$
- B. $55x + 25y = 160$
 $x + y = 4$
- C. $25x + 55y = 4$
 $x + y = 160$
- D. $25x + 55y = 160$
 $x + y = 4$

$$y = 6x + 3$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which equation could be the second equation in this system?

- A. $y = 2(6x) + 3$
- B. $y = 2(6x + 3)$
- C. $2(y) = 2(6x) + 3$
- D. $2(y) = 2(6x + 3)$